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2014.0 RANGE ROVER (LG), 310-01

FUEL TANK AND LINES - V8 N/A 5.0L PETROL/V8 S/C 5.0L PETROL

PRINCIPLE OF OPERATION

For a detailed description of the fuel tank and lines system and operation, refer to the relevant Description and Operation section of the workshop manual.

REFER to: Fuel Tank and Lines (310-01 Fuel Tank and Lines - V8 5.0L Petrol/V8 S/C 5.0 L Petrol, Description and Operation).

INSPECTION AND VERIFICATION



WARNINGS:

- Do **NOT** carry out any work on the fuel system with the engine running. The fuel pressure within the system can be as high as 1600 bar (23,206 lb/in²). Failure to follow this instruction may result in personal injury.
- Eye protection must be worn at all times when working on or near any fuel related components. Failure to follow this instruction may result in personal injury.
- This procedure involves fuel handling. Be prepared for fuel spillage at all times and always observe fuel handling precautions. Failure to follow this instruction may result in personal injury.
- After carrying out repairs, the fuel system must be checked visually for leaks. This should be done after the engine has been run, but with the engine switched **OFF** . Failure to follow this instruction may result in personal injury.
- If taken internally, **DO NOT** induce vomiting. Seek immediate medical attention. Failure to follow this instruction may result in personal injury.
- If fuel contacts the eyes, flush the eyes with cold water or eyewash solution and seek medical attention. Failure to follow this instruction may result in personal injury.
- Wash hands thoroughly after handling, as prolonged contact may cause irritation. Should irritation develop, seek medical attention. Failure to follow this instruction may result in personal injury.



CAUTIONS:

- Before disconnecting any part of the system, it is imperative that all dust, dirt and debris is removed from around components to prevent ingress of foreign matter into the fuel system. Failure to follow this instruction may result in damage to the vehicle.
- The fuel pipes between the injectors and the rail must be discarded after each use, and new pipes installed. Failure to follow this instruction may result in damage to the vehicle.
- It is essential that absolute cleanliness is observed when working with these components. Always install blanking plugs to any open orifices or lines. Failure to follow this instruction may result in damage to the vehicle.
- Make sure that the workshop area in which the vehicle is being worked on is as clean and dust-free as possible. Areas in which work on clutches, brakes or where welding or machining are carried out are not suitable in view of the risk of contamination to the fuel system. Failure to follow this instruction may result in damage to the vehicle.
- Make sure that any protective clothing worn is clean and made from lint-free non-flocking material. Failure to follow this instruction may result in damage to the vehicle.
- Make sure that any protective gloves worn are new and are of the non-powdered latex type. Failure to follow this instruction may result in damage to the vehicle.
- Make sure that clean, non-plated tools are used. Clean tools using a new brush that will not lose its bristles and fresh cleaning fluid prior to starting work on the vehicle. Failure to follow this instruction may result in damage to the vehicle.
- Use a steel-topped work bench and cover it with clean, lint-free, non-flocking material. Failure to follow this instruction may result in damage to the vehicle.
- Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle.

NOTES:

- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- When measuring fuel sender resistance values with a multimeter, it is critical to use the correct multimeter setting. The multimeter should **not** be on the 'Auto' setting and **must** be set to 'Manual'. This will help prevent incorrect diagnosis and unnecessary replacement of fuel senders. If the multimeter range is set at 'Auto' then, during a sweep of the sender from 50 Ohms to

998 Ohms, the multimeter has to change its measurement range. For approximately 1 second, during the range switch over point, the multimeter display indicates an open circuit. This can lead to a mis-diagnosis of a fuel sender fault.

1. Verify the customer concern.

1. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Fuel level ▪ Contaminated fuel ▪ Fuel supply line(s) ▪ Fuel return line(s) ▪ High-pressure fuel supply line(s) ▪ Fuel tank filler pipe ▪ Fuel leak(s) ▪ Fuel tank ▪ Fuel filler cap ▪ Fuel filter ▪ Push connect fittings ▪ Fuel rail ▪ Fuel injection pump ▪ Exhaust gas recirculation system	▪ Battery charge and condition ▪ Fuse(s) ▪ Fuel pump module (lift pump) ▪ Fuel pump module relay ▪ Electrical connector(s) ▪ Damaged or corroded wiring harness ▪ Fuel volume control valve ▪ Fuel pressure control valve ▪ Engine control module

1. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

1. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC index.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Engine cranks, but does not fire	▪ Engine breather system	Ensure the engine breather system is free from restriction and is correctly installed. Check for ignition system, fuel system and electronic engine control DTCs and refer to the

	disconnected /restricted ▪ Ignition system ▪ Fuel system ▪ Electronic engine control	relevant DTC index
Engine cranks and fires, but will not start	▪ Evaporative emissions purge valve ▪ Fuel pump ▪ Spark plugs ▪ HT short to ground (tracking) check rubber boots for cracks /damage ▪ Ignition system	Check for evaporative emissions, fuel system and ignition system related DTCs and refer to the relevant DTC index
Difficult cold start	▪ Engine coolant level /anti-freeze content ▪ Battery ▪ Electronic engine controls ▪ Exhaust gas recirculation valve stuck open ▪ Fuel pump ▪ Purge valve	Check the engine coolant level and condition. Ensure the battery is in a fully charged and serviceable condition. Check for electronic engine controls, engine emissions, fuel system and evaporative emissions system related DTCs and refer to the relevant DTC index
Difficult hot start	▪ Injector leak ▪ Electronic engine control ▪ Purge valve ▪ Fuel pump ▪ Ignition system ▪ Exhaust gas recirculation valve stuck open	Check for injector leak, install new injector as required. Check for electronic engine controls, evaporative emissions, fuel system, ignition system and engine emission system related DTCs and refer to the relevant DTC index

Difficult to start after hot soak (vehicle standing, engine off, after engine has reached operating temperature)	▪ Injector leak ▪ Electronic engine control ▪ Purge valve ▪ Fuel pump ▪ Ignition system ▪ Exhaust gas recirculation valve stuck open	Check for injector leak, install new injector as required. Check for electronic engine controls, evaporative emissions, fuel system, ignition system and engine emission system related DTCs and refer to the relevant DTC index
Engine stalls soon after start	▪ Breather system disconnected /restricted ▪ Engine control module relay ▪ Electronic engine control ▪ Ignition system ▪ Air intake system restricted ▪ Air leakage ▪ Fuel lines	Ensure the engine breather system is free from restriction and is correctly installed. Check for electronic engine control, ignition system and fuel system related DTCs and refer to the relevant DTC index. Check for blockage in air filter element and air intake system. Check for air leakage in air intake system
Engine hesitates /poor acceleration	▪ Fuel pressure, fuel pump, fuel lines ▪ Injector leak ▪ Air leakage ▪ Electronic engine control ▪ Throttle motor ▪ Restricted accelerator pedal travel (carpet, etc) ▪ Ignition system ▪ Exhaust gas recirculation valve stuck open	Check for fuel system related DTCs and refer to the relevant DTC index. Check for injector leak, install new injector as required. Check for air leakage in air intake system. Ensure accelerator pedal is free from restriction. Check for electronic engine controls, ignition, engine emission system and transmission related DTCs and refer to the relevant DTC index

	▪ Transmission malfunction	
Engine backfires	▪ Fuel pump /lines ▪ Air leakage ▪ Electronic engine controls ▪ Ignition system ▪ Sticking variable camshaft timing hub	Check for fuel system failures. Check for air leakage in intake air system. Check for electronic engine controls, ignition system and variable camshaft timing system related DTCs and refer to the relevant DTC index
Engine surges	▪ Fuel pump /lines ▪ Electronic engine controls ▪ Throttle motor ▪ Ignition system	Check for fuel system failures. Check for electronic engine controls, throttle system and ignition system related DTCs and refer to the relevant DTC index
Engine detonates /knocks	▪ Fuel pump /lines ▪ Air leakage ▪ Electronic engine controls ▪ Sticking variable camshaft timing hub	Check for fuel system failures. Check for air leakage in intake air system. Check for electronic engine controls and variable camshaft timing system related DTCs and refer to the relevant DTC index
No throttle response	▪ Electronic engine controls ▪ Throttle motor	Check for electronic engine controls and throttle system related DTCs and refer to the relevant DTC index
Poor throttle response	▪ Breather system disconnected /restricted ▪ Electronic engine controls ▪ Transmission malfunction	Ensure the engine breather system is free from restriction and is correctly installed. Check for electronic engine controls, transmission and traction control related DTCs and refer to the related DTC index. Check for air leakage in intake air system

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| | <ul style="list-style-type: none">▪ Traction control event▪ Air leakage | |
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DTC INDEX

For a complete list of all Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: Diagnostic Trouble Code (DTC) Index - V8 5.0L Petrol/V8 S/C 5.0L Petrol, DTC: Engine Control Module (ECM) (100-00 General Information, Description and Operation).